NEXUS-1

PROJECT ROADMAP — v2

From Phase-0 demonstrator to an industrial-grade platform

Staged plan with work-hour estimates · .NET C# microservices · REST · SignalR · Angular · Kafka

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ESTIMATE, NOT A QUOTE. Hours are planning estimates for senior engineers and depend on team, seniority and final scope. Scope = Target A (industrial platform). The regulated-deployment path (Track B) is a separate, specialist-led, non-engineering track and is not estimated in hours.

WHAT CHANGED FROM v1. Adds the Training & Drills service to the decomposition; adds Phase P8 (Hybrid Digital Twin & advisory-on-live — previously the destination but unpriced); adds the DSLM assistant to the analytics tier; records the binding technology decisions (Kafka-only, no Orleans, PostgreSQL/TimescaleDB) that supersede earlier document references; adds an operating-mode capability map, milestone exit gates and a risk register. Phase numbering: Production Readiness is now P6, Protocol Integration P7, Hybrid Twin P8.

How to read this roadmap

This plan takes NEXUS-1 from its current state — a complete client-side demonstrator (≈41 screens across ≈25 menu areas, the six-group point-kinetics engine with Doppler/moderator/xenon feedback, cross-module event behaviour, Training Mode with scored drills; but no backend, persistence, auth, real-time transport, tests or infrastructure) — to an industrial-grade platform: .NET C# microservices exposed through REST, an Angular front end, SignalR for real-time push, and Kafka as the single messaging substrate for events, commands and telemetry. Most of the work is not re-skinning the tabs; it is building the distributed system the tabs sit on top of.

Scope assumed

- **Target A — industrial platform.** Productionised monitoring / analytics / digital-twin / decision-support layers, validated against simulation and historian data, integration-ready. NOT the regulated, plant-connected path — that is Track B, covered separately and not estimated as engineering.
- **Architecture — microservices, started coarse-grained.** Independently deployable services aligned to the bounded contexts on the next page, asynchronous over Kafka, synchronous only through the API gateway, database-per-service. Several services begin life merged and split only when load or team topology justifies it — premature splitting is the main way this estimate overruns.
- **Team.** A small team of senior engineers (backend / frontend / platform-DevOps, with data/ML and part-time architecture). All figures are senior-engineer hours.
- **'Complete'.** Every current console tab taken to production grade, sequenced vertical-slice-first (milestone M1) so de-risking and demonstrable value come early.

How the hours work

- Each task carries a **Low** and **High** estimate of focused engineering effort (design + build + test + review).
- The **Planning** figure per phase is the midpoint plus ≈15% contingency — use this for budgeting.
- A separate line covers program management & technical leadership across all phases.
- Calendar time assumes ≈6 productive hours per engineer per day (≈126/month); coordination, dependencies and ramp-up add overhead beyond pure division.
- Excluded: hardware/licences, third-party services, and the regulatory track (Track B).
- **Every phase estimate is re-baselined at the end of M1** — the vertical slice converts these ranges into measured velocity.

Technology decision records (binding for this plan)

These decisions resolve inconsistencies between earlier NEXUS-1 documents. Where a v1 document says otherwise, this table governs.

ID	Decision	Rationale / consequence
D1	Messaging: **Kafka only** — events, commands and telemetry on one broker.	Supersedes the Azure Service Bus + Event Hubs reference in the Project Brief. One broker, one operational skillset; the replayable log doubles as the event-sourcing / twin-replay substrate. Single-producer today is deliberate future-proofing for multi-source ingest (P7).
D2	Compute: **no Orleans.** Stateful twin via partitioned worker services.	Kafka partition affinity + in-memory state + snapshot/restore + rebalance protocol, hand-built. Cheaper conceptually, costlier in engineering — priced explicitly in P1.2 and P8.2 and spiked at M1.
D3	Persistence: **PostgreSQL per service; TimescaleDB historian.**	Supersedes the SQL Server reference. No per-core licensing, first-class time-series (compression, continuous aggregates), air-gap friendly. EF Core keeps SQL Server as a swap-in if a customer's licensing dictates.
D4	Real-time: **SignalR with Redis backplane.**	Horizontal fan-out of telemetry/presence streams across gateway replicas; backpressure tested in P0.9/P6.2.
D5	Edge: **YARP gateway + BFF; OpenAPI-first.**	Versioned per-service contracts; Angular client generated from OpenAPI — no hand-written HTTP layer.
D6	Identity: **Keycloak (OIDC/OAuth2), RBAC.**	Self-hosted, air-gap deployable, no cloud dependency; sessions, tokens, service accounts.
D7	Contracts: **Avro + schema registry; Pact in CI.**	Versioned event/telemetry schemas; consumer-driven contract tests gate every service build.
D8	Frontend: **Angular** (Signals; NgRx where state warrants), three.js retained for 3D.	Mirrors the existing menu; design tokens ported from the Phase-0 CSS.

ID	Decision	Rationale / consequence
D9	Observability: **OpenTelemetry → Prometheus → Grafana**, logs via Loki.	Tracing/metrics/logging baked into the P0.11 service template, not retrofitted.
D10	Deploy: **Docker/Kubernetes, Helm**, air-gapped install supported.	Offline registry mirror + offline charts — required for the eventual plant-adjacent posture.
D11	AI tier: **local Ollama DSLM, RAG-grounded, advisory only.**	Nuclear-domain assistant served air-gapped; answers cite the registry, incidents and documents. Never wired to control or safety actuation.

Architecture: microservices decomposition

Each service owns its data, publishes domain events to Kafka, and is reached synchronously only through the API gateway. Cross-cutting concerns (gateway, identity, observability) are shared platform, not per-service. **Change from v1:** the Training & Drills service is added — Training Mode is a live, scored capability of the Phase-0 console and one of the six operating modes, and it was missing from the v1 decomposition.

Service	Responsibility	Backs (tabs / mode)
Identity & Access	AuthN/Z, RBAC, sessions, tokens	(all — security)
API Gateway / BFF	Edge routing, aggregation, auth enforcement	(all)
Realtime / Notification	SignalR fan-out, presence, push	(all live screens)
Simulation / Digital Twin	Physics, point-kinetics, fleet state, twin	Overview, Reactor, Kinetics, 3D, Twin
Telemetry & Historian	Ingest, time-series store, queries	Trends, Power, Coolant, Steam
Command	Rod / SCRAM control path, validation	Control Rods
Training & Drills (new)	Drill scenarios, scoring, session records, instructor view	Training Mode
Alarm	Alarm rules, dedup, ack, escalation	Alarms & Events
Audit	Hash-chained tamper-evident action log	Compliance / Audit
Asset & Health	Component registry, live health, aging	Component Registry, Aging
Inspection	NDT records and inspection workflow	Rod Inspection
Personnel	Staffing, rosters, coverage rules	Personnel
Robotics	Robot / vehicle fleet, mission readiness	Robotics & Vehicles
Zone Access	Live presence, permissions matrix	Zone Access
Incident & Root-cause	Event store, timeline, causal analysis	Incident, Root Cause Graph
Analytics / AI	Predictive maintenance, dependency/causal, DSLM assistant	AI Diagnostics, Dependencies
RL Optimiser	Twin-only advisory optimisation sandbox	Optimization (RL)
Protocol Gateways (P7)	OPC-UA / MQTT / Modbus, read-only ingest	(integration, optional)
Assimilation / Hybrid (P8)	State estimation, live-sim reconciliation, branchable twin	Digital Twin, Advisory

Start coarse-grained. Sensible initial merges: Asset & Health + Inspection; Analytics + RL; Training inside Simulation; Personnel + Zone Access. Split only when load or team structure demands it.

Operating-mode capability map

NEXUS-1's destination is all six operating modes. This table states exactly which engineering work unlocks each mode, and what each mode is gated on. It is the contract between this roadmap and the product's mode taxonomy.

Mode	Unlocked by	Milestone	Gated on
Simulation	Server-side sim tier: P0–P2 (Kafka spine, sim service, Angular console)	M2	— (active in Phase 0, client-side)
Training	P3 Training & Drills service + scoring parity with Phase-0 drills	M3	M2
Shadow	P7 read-only protocol gateways + divergence dashboard	M6	M5 + an available plant/historian data source
Advisory	P4 analytics tier operating on live data, P8.7 guardrails & reproducibility	M7 (first stage)	M6; Track B engagement for any on-site use
Supervisory	P5 security/OT posture + organisational integration	beyond M7	Track B progress; customer OT governance
Hybrid Digital Twin	P8 data assimilation + branchable what-if off live state	M7	M6; hardest phase algorithmically

No control authority — invariant across all modes. The platform watches, trains, advises and reconciles. It never actuates plant equipment; certified reactor-protection functions stay hardware-isolated, deterministic and outside this system. Shadow and beyond are read-only by network policy and by code audit (gate M6), not by convention.

Executive summary

Phase	Stage	Low	High	Planning
P0	Foundations & Architecture	345	595	540
P1	Backend Core, Simulation Port & Real-time Spine	730	1,200	1,110
P2	Angular Frontend Rebuild	630	1,090	990
P3	Domain Services (incl. Training & Drills)	740	1,230	1,130
P4	AI / Analytics Services (incl. DSLM assistant)	520	880	800
P5	Security & OT Hardening	380	660	600
P6	Production Readiness & QA	630	1,110	1,000
PM	Program management & technical leadership (all phases)	520	820	770
	CORE PLATFORM TOTAL (Target A)	**4,495**	**7,585**	**≈6,950**
P7	Protocol Integration — optional, shadow mode	400	720	640
P8	Hybrid Digital Twin & Advisory-on-live — optional, gated	480	840	760
	TOTAL — all six modes engineered	**5,375**	**9,145**	**≈8,350**

All figures in engineering work-hours. Planning = midpoint + ≈15% contingency. P7/P8 are real cost but deferrable: they are only meaningful once a live or historian data source exists.

Indicative calendar (planning hours)

Team	Composition	Naive	Realistic
Solo	1 senior full-stack	≈55 mo core	≈4+ yr — **not advised** (breadth, zero parallelism, bus factor)
Small (3)	backend / frontend / platform-DevOps	≈18 mo core	≈15–18 months to a hardened core v1; P7 adds 4–6 mo (parallelisable); P8 adds 4–7 mo after a data source exists
Balanced (5–6)	2 BE, 2 FE, 1 SRE, 1 data/ML, p/t architect+QA	≈11 mo core	≈10–13 months to a strong core v1; P5/P6 run in parallel with P3/P4; P7/P8 partly parallel

Realistic figures add coordination overhead, inter-phase dependencies and ramp-up.

Milestones & sequencing

Do not build all tabs at once. One vertical slice through the entire stack first, then fan out the remaining domains (which mostly repeat established patterns).

Milestone	Content	Approx. hours
M1 — Vertical slice	P0 essentials + minimal P1 (sim service, gateway, SignalR, Kafka, historian) + P2 shell (Overview, Fleet, live Reactor) + Alarms + Audit + auth. Includes the D2 stateful-compute spike. Proves the hard parts.	≈750–950
M2 — Simulation tier	Remaining P1 + P2: full reactor, kinetics, 3D, steam/coolant, trends on real data. Unlocks server-side **Simulation Mode**.	≈1,300–1,700
M3 — Domains complete	All P3 services and screens backed by real data and workflow, including Training & Drills. Unlocks **Training Mode**.	≈750–1,250
M4 — Analytics + security	P4 analytics tier (incl. DSLM) and P5 hardening, largely parallel.	≈900–1,540
M5 — Production v1	P6 complete: tested, observable, resilient, deployed on Kubernetes.	≈540–940
M6 — Shadow integration (opt.)	P7 read-only protocol gateways + divergence dashboard. Unlocks **Shadow Mode**.	≈400–720
M7 — Hybrid twin (opt.)	P8 assimilation, branchable twin, advisory-on-live guardrails. Unlocks **Hybrid** and the live stage of **Advisory**.	≈480–840

M1 is the most important decision in this plan: ~3–4 months for a small team yields a demonstrably industrial end-to-end system before a year of spend, and converts every estimate below into measured velocity.

Milestone exit gates (definition of done)

A milestone is not 'done when the tasks are done'; it is done when its gate passes. Gates are the verifiable claims this roadmap makes — the same honesty discipline the Phase-0 console applies to its displayed numbers, applied to the build itself.

Gate	Exit criteria
M1	End-to-end on a dev K8s cluster: login → live Reactor screen at ≥1 Hz over SignalR, sourced from the server sim via Kafka and persisted to the historian; an alarm raised, escalated and acknowledged; the audit chain cryptographically verifies; CI builds/tests/containers every service. D2 spike result recorded as an ADR. **All phase estimates re-baselined from measured velocity.**
M2	Golden-master parity: the server-side kinetics reproduces the Phase-0 browser solver within stated tolerance on all five drill scenarios and the closed-form references from Model Analysis. 24 h soak with no state drift or memory growth. Historian queries meet their latency SLA at representative cardinality.
M3	Every console tab backed by persisted data and workflow; drill scoring reproduces Phase-0 results; all domain events on Kafka with Pact contract tests green; the tamper-evident audit survives an adversarial tamper attempt in test.
M4	External pen test complete with no open critical/high findings; model registry, evaluation and drift monitoring live; every advisory output carries guardrail labelling and provenance; DSLM answers are retrieval-grounded with citations, and refuse when sources are absent.
M5	Chaos/failover drills pass (broker loss, pod kill, network partition); HA verified; DR restore rehearsed end-to-end including air-gapped install; runbooks complete; SLOs defined and met for the real-time pipeline.
M6	Read-only OPC-UA ingest from a reference source (simulator-as-source rig if no plant data); divergence dashboard live; **absence of any command path proven** by network policy and a code audit, not asserted.
M7	Assimilation holds the twin within stated tolerance against replayed historian data; a what-if branch forks off live state in bounded time, runs fast-time, and merges or discards cleanly; every live advisory is reproducible from logged inputs.

Detailed stages (work breakdown)

P0 — Foundations & Architecture

Stand up the skeleton everything else hangs from: architecture, environments, pipelines and the auth model. No business features yet — this phase exists to de-risk the build.

ID	Task	Low	High	Notes
P0.1	Microservices architecture, ADRs, service decomposition	55	90	Service boundaries, ownership, Kafka topic design
P0.2	.NET solution scaffolding + shared libraries	20	35	Projects, shared kernel
P0.3	Angular workspace + design-system scaffold	25	40	Port the existing CSS tokens
P0.4	Local dev environment (docker-compose)	30	50	Kafka, Postgres, TimescaleDB, Redis, Keycloak
P0.5	CI/CD pipeline skeleton (per service)	35	60	Build / test / lint / SCA / containerize
P0.6	IaC skeleton (Terraform / Helm)	30	55	Baseline cluster + environments
P0.7	Identity / auth + RBAC model (Keycloak)	25	45	Realms, roles, service accounts
P0.8	Event & telemetry contracts + schema registry	30	50	Avro, versioning policy
P0.9	Real-time throughput spike (SignalR + Kafka + Redis)	25	45	Prove the spine early
P0.10	Standards, repo, branching, board setup	15	25	Conventions & templates
P0.11	Service template / golden-path scaffold	25	45	Health, logging, tracing, OTel baked in
P0.12	API gateway + service discovery + central config	30	55	YARP, K8s DNS, config service
	Phase subtotal	**345**	**595**	**Planning ≈540 h**

› Highest leverage of the whole programme — a weak P0 multiplies cost everywhere downstream. Output: a running empty skeleton that builds, deploys and authenticates.

P1 — Backend Core, Simulation Port & Real-time Spine

The heart of the platform: the server-side simulation/twin, the API, and the SignalR + Kafka + historian pipeline streaming live state to clients.

ID	Task	Low	High	Notes
P1.1	Core domain model + persistence (EF Core / Postgres)	50	80	Units, components, events
P1.2	Simulation/Twin service — fleet model, master tick, partitioned state	70	110	D2 pattern: partition affinity, snapshot/restore
P1.3	Point-kinetics engine port + unit tests	90	140	**HIGH RISK** — 6 delayed groups, Doppler/moderator/xenon
P1.4	Thermal-hydraulics, secondary & grid models	50	80	Coolant / steam / power
P1.5	Deterministic sim test harness (golden master)	40	70	Parity vs Phase-0 solver + closed-form refs
P1.6	Edge API gateway + BFF + per-service contracts	60	95	Versioned, OpenAPI, aggregation
P1.7	SignalR hubs + Redis backplane + backpressure	50	80	Telemetry / presence streams
P1.8	Kafka integration (topics, registry, idempotency)	70	110	Producers / consumers, exactly-once where needed
P1.9	TimescaleDB historian write path + retention	50	80	Downsampling, continuous aggregates
P1.10	Command path (rod / SCRAM) + audit emission	40	70	Consequential actions, validation
P1.11	Read models, caching, health checks	30	50	Query side
P1.12	API error model, configuration, feature flags	40	80	Cross-cutting
P1.13	Inter-service comms + consistency (outbox / saga)	60	100	Async events + sync gRPC/REST
P1.14	Per-service data ownership + migrations	30	55	DB-per-service, no shared schema
	Phase subtotal	**730**	**1,200**	**Planning ≈1,110 h**

› The point-kinetics port (P1.3) is the single biggest technical risk — protect its schedule and test it against the golden master, hard. Output: live, persisted plant state streaming to a client over SignalR.

P2 — Angular Frontend Rebuild

Re-implement the console as a proper Angular application driven by the API and the real-time stream, mirroring the existing menu, with the visualization-heavy screens rebuilt.

ID	Task	Low	High	Notes
P2.1	App shell, routing (menu mirror), theming	60	100	Design-system port
P2.2	Auth (OIDC), route guards, RBAC-aware UI	40	70	
P2.3	Generated API client + SignalR client / reconnect	40	70	From OpenAPI (D5)
P2.4	State management + real-time store	50	90	Signals / NgRx
P2.5	Simulation screens: Overview / Fleet / Power	60	100	
P2.6	Reactor screens: Core / Rods / Neutronics / Steam / Coolant	80	130	Interactive controls
P2.7	Kinetics screen (charts + controls)	40	70	
P2.8	3D screens (Reactor 3D + Plant 3D)	90	150	three.js, performance
P2.9	Trends / History (time-series)	40	70	
P2.10	Shared component library, a11y, responsive	50	90	
P2.11	i18n scaffold (EN / EL)	20	40	
P2.12	Frontend tests (unit / component / e2e)	60	110	Playwright / Cypress
	Phase subtotal	**630**	**1,090**	**Planning ≈990 h**

› Output: the simulation tier of the console, production-grade, on real data — unlocks server-side Simulation Mode (gate M2).

P3 — Domain Services (incl. Training & Drills)

Turn the record-keeping and workflow tabs into real services with persistence and process — and give Training Mode, one of the six operating modes, a first-class service of its own.

ID	Task	Low	High	Notes
P3.1	Alarm engine + ack / escalation workflow + UI	90	140	Dedup, severity, routing — larger than it looks
P3.2	Tamper-evident audit (hash-chained, append-only) + UI	70	110	Real crypto + verification — larger than it looks
P3.3	Asset / component registry + live health + UI	70	110	Per-unit equipment
P3.4	Aging & degradation service + UI	50	80	
P3.5	Inspection / NDT records + workflow + UI	60	100	
P3.6	Personnel / staffing + coverage rules + UI	60	100	
P3.7	Robotics & vehicles + mission readiness + UI	60	100	
P3.8	Zone access (presence + permissions matrix) + UI	50	90	
P3.9	Incident analysis (event store + timeline) + UI	50	90	
P3.10	Root-cause graph service + ranking + UI	70	120	Causal backbone, alarm-flood collapse
P3.11	Domain events on Kafka + projections	50	90	
P3.12	**Training & Drills service** + scoring + instructor view	60	100	**New in v2.** Scenarios, session records, scoring parity with Phase-0
	Phase subtotal	**740**	**1,230**	**Planning ≈1,130 h**

› Output: every non-simulation tab backed by a real data model and workflow; Training Mode unlocked (gate M3).

P4 — AI / Analytics Services (incl. DSLM assistant)

The analytics tier on top of the historian: predictive maintenance, the dependency/causal model, the RL optimiser kept strictly as a twin-only advisory sandbox — and the DSLM assistant, previously documented as a roadmap layer but unpriced.

ID	Task	Low	High	Notes
P4.1	Feature pipeline off the historian	60	110	TimescaleDB → features
P4.2	Predictive-maintenance baselines + model serving	90	150	Statistical first, ML once data exists
P4.3	Dependency / causal service	60	100	Graph compute, weights
P4.4	RL optimiser (twin-only, advisory)	90	150	Sandbox + readable policy
P4.5	Model registry, evaluation, drift monitoring	50	90	
P4.6	Advisory UI surfacing + guardrail labelling	40	70	
P4.7	**DSLM assistant** — Ollama serving, RAG over registry/incidents/docs, eval harness	130	210	**New in v2.** Air-gapped, citation-grounded, refuses without sources; advisory only
	Phase subtotal	**520**	**880**	**Planning ≈800 h**

› Realism here is bound by data — start transparent/heuristic, become ML once historian data exists. Advisory only: never wired to control or safety actuation.

P5 — Security & OT Hardening

Critical-infrastructure security posture: identity, service-to-service trust, secrets, segmentation, supply-chain scanning and an independent penetration test.

ID	Task	Low	High	Notes
P5.1	AuthN/Z hardening + RBAC enforcement tests	60	100	
P5.2	mTLS / service identity, secrets (KMS), HSM	70	120	
P5.3	Segmentation / zero-trust, gateway hardening	50	90	
P5.4	Security audit coverage + SIEM hooks	40	70	
P5.5	Threat model + SCA / SAST / DAST in CI	50	90	

ID	Task	Low	High	Notes
P5.6	External penetration test + remediation	80	140	Third-party, near feature-complete
P5.7	OT security policy pack + documentation	30	50	IEC 62645-informed posture
	Phase subtotal	**380**	**660**	**Planning ≈600 h**

› Several items run in parallel with P3/P4.

P6 — Production Readiness & QA

Operable and trustworthy in production: the full test pyramid, load/soak/chaos testing of the real-time pipeline, observability, HA/DR and a Kubernetes production deployment. Runs largely in parallel with P3–P5. (Numbered P7 in v1.)

ID	Task	Low	High	Notes
P6.1	Test strategy + pyramid (unit / integration / contract)	90	150	
P6.2	Load & soak testing (pipeline / Kafka / SignalR fan-out)	70	120	
P6.3	Chaos / resilience + failover drills	50	90	
P6.4	Observability (OpenTelemetry, dashboards, alerting)	70	120	Prometheus / Grafana / Loki
P6.5	HA / redundancy + backpressure tuning	60	100	
P6.6	Kubernetes production deploy (Helm, autoscaling)	80	140	
P6.7	DR + backup / restore + air-gapped support	60	110	
P6.8	Documentation, runbooks, ops, API reference	60	110	
P6.9	Service mesh + resilience patterns	50	90	Circuit breakers, retries, bulkheads, timeouts
P6.10	Consumer-driven contract testing (Pact)	40	80	Cross-service contracts enforced in CI
	Phase subtotal	**630**	**1,110**	**Planning ≈1,000 h**

› Output: a deployable, observable, resilient v1 of the platform (gate M5).

P7 — Protocol Integration (optional, shadow mode)

Read-only integration with plant data through isolated gateway services. Deferrable; only meaningful once a data source is available. A simulator-as-source rig stands in until then. (Numbered P6 in v1.)

ID	Task	Low	High	Notes
P7.1	Gateway architecture + isolation	40	70	Separate network zone, one-way flow
P7.2	OPC-UA gateway	80	140	
P7.3	MQTT gateway	50	90	
P7.4	Modbus TCP gateway	50	90	
P7.5	Normalization / validation to canonical model	60	110	
P7.6	Read-only twin sync to live data (divergence)	70	130	Shadow mode
P7.7	Conformance / integration test rig	50	90	Simulator-as-source until plant data exists
	Phase subtotal	**400**	**720**	**Planning ≈640 h**

› Strictly read-only; no command path to plant equipment — proven at gate M6, not asserted.

P8 — Hybrid Digital Twin & Advisory-on-live (optional, gated) — NEW

The hardest phase algorithmically, and the one v1 named as the destination but did not price. Keeps the simulation continuously in step with live (or replayed historian) data, supports commandable what-if branches off the live state, and puts guardrails under advisory outputs computed from live data. Gated on P7 and a data source.

ID	Task	Low	High	Notes
P8.1	State estimation / data assimilation on the reduced model	90	150	**HIGH RISK** — EnKF/UKF-class filter over kinetics + TH state
P8.2	Stateful twin compute pattern (no Orleans)	80	160	Partitioned workers, snapshot/restore, rebalance, failover — D2 made real
P8.3	Live–sim reconciliation engine + divergence classification	60	100	Residuals → sensor fault vs model error vs plant change
P8.4	Branchable what-if off live state	70	120	Fork, fast-time run, merge/discard; bounded fork time
P8.5	Model recalibration loop against live data	70	120	Parameter updates with provenance, never silent
P8.6	Time alignment, latency & data-quality handling	50	90	Clock skew, gaps, late/duplicate samples
P8.7	Advisory-on-live guardrails, reproducibility & logging	60	100	Every recommendation replayable from logged inputs

ID	Task	Low	High	Notes
	Phase subtotal	**480**	**840**	**Planning ≈760 h**

› Output: Hybrid Digital Twin Mode and the live stage of Advisory Mode (gate M7). Validated first against replayed historian data — a real-time plant feed is not a prerequisite for building and proving the algorithms.

Separate Track B — Nuclear V&V / Regulated Deployment

Connecting the platform to, and having it trusted by, a real nuclear plant is a different undertaking from the engineering above. It is largely organisational and specialist-led, and it is deliberately not estimated in engineering hours — doing so honestly requires a nuclear QA organisation, not a software team. Indicative activities:

- Formal verification & validation of the simulation/twin models against qualified references.
- Requirements traceability and a documented safety / security case.
- Alignment with the relevant standards — IEC 61513 (system level), IEC 60880 (software), IEC 62645 (cyber-security).
- Independent verification and assessment by a separate competent party.
- Regulator engagement and licensing-basis documentation.
- Even then, the platform remains in the monitoring / analytics / advisory layer; certified reactor-protection functions stay hardware-isolated, deterministic and outside this system.

Indicative duration: 12–36+ months, specialist-led, partly in parallel with later engineering phases. Treat it as a programme, not a sprint. The realistic first regulated step is a read-only Advisory deployment outside the safety-classified I&C envelope, with organisational backing — never a solo effort.

Risk register

ID	Risk	Lik.	Imp.	Mitigation
R1	Point-kinetics port fidelity (P1.3)	M	H	Golden-master harness vs Phase-0 solver and closed-form references; protect the schedule; no parallel work stacked on it
R2	Real-time pipeline throughput (Kafka + SignalR fan-out)	M	H	P0.9 spike before commitment; Redis backplane; load/soak in P6.2 with explicit SLOs
R3	Stateful twin without Orleans (D2 / P8.2)	M	H	Prototype partition-affinity + snapshot/restore in the M1 spike; documented fallback: single-writer sim service, scale by fleet sharding
R4	Over-decomposition of services	M	M	Start coarse-grained (merges listed under the decomposition); split only on measured load or team topology
R5	Alarm engine and cryptographic audit underestimated	M	M	Treat P3.1/P3.2 as mini-projects with their own design reviews
R6	No live data source for P7/P8	H	M	Defer both phases; validate algorithms against replayed historian datasets and the simulator-as-source rig (P7.7)
R7	Estimate validity	H	M	M1 converts ranges into measured velocity; re-baseline at every milestone gate
R8	Single-architect bus factor	M	H	ADRs, golden-path service template and runbooks from P0; pair onboarding at M1; documentation treated as load-bearing
R9	Security regressions discovered late	L	H	SCA/SAST/DAST in CI from P0.5; threat model early (P5.5); external pen test at feature-complete
R10	Scope creep from the 41-screen surface	M	M	Vertical-slice sequencing; new screens enter only through the decomposition table and a priced task

Assumptions, caveats & authorship

- Planning estimates, not a fixed-price quotation. Actuals depend on team seniority, agreed scope, and the open scoping questions (team size/mix, target depth, definition of 'complete').
- The point-kinetics port (P1.3) and data assimilation (P8.1) are the dominant technical risks; the alarm engine, cryptographic audit and the real-time pipeline are the next most uncertain.
- Kafka with a single simulation producer is future-proofing for multi-source ingestion (P7), not a present functional need; it adds operational weight early but suits the architecture and integration goals.
- Microservices add platform, contract-testing and coordination overhead (P0.11–12, P1.13–14, P6.9–10, and the PM line). Over-decomposition is the main way this estimate would overrun.
- Document reconciliation: this roadmap supersedes the v1 roadmap and corrects the Project Brief's stack references (Kafka-only replaces Azure Service Bus + Event Hubs; PostgreSQL/TimescaleDB replaces SQL

Server) and the screen count (≈41 screens across ≈25 menu areas).

• Re-validate every phase estimate at the end of Milestone M1 — the vertical slice converts these ranges into measured velocity.